Practical 32

Aim: To implement Transpose of a Matrix using C and Java.

Theory: If matrix **A** is of size m × n, then its transpose $\mathbf{A^T}$ is of size n × m.

Transpose means:

A[i][j] → A[j][i]

If a matrix **A** has size **m × n**, then its transpose $\mathbf{A^T}$ will have size **n × m**.

That means **rows become columns and columns become rows**.

Procedure / Algorithm for Transpose of a Matrix

Step 1: Start the program.

Step 2: Read the number of rows (m) and columns (n).

Step 3: Declare a matrix of size m × n.

Step 4: Input elements into the matrix.

Step 5: Declare another matrix of size n × m (for transpose).

Step 6: Use nested loops:

- Outer loop for rows (i = 0 to m-1)
- Inner loop for columns (j = 0 to n-1)
- Assign:

transpose[j][i] = matrix[i][j];

Step 7: Display the original matrix.

Step 8: Display the transpose matrix.

Step 9: Stop the program.

C Program:

```
#include <stdio.h>

int main() {
    int rows, cols;

    printf("Enter number of rows: ");
    scanf("%d", &rows);

    printf("Enter number of columns: ");
    scanf("%d", &cols);

    int matrix[rows][cols];
    int transpose[cols][rows];

    // Input matrix
    printf("Enter elements of matrix:\n");
    for(int i = 0; i < rows; i++) {
        for(int j = 0; j < cols; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    // Transpose logic
    for(int i = 0; i < rows; i++) {
        for(int j = 0; j < cols; j++) {
```

```
            transpose[j][i] = matrix[i][j];
        }
    }

    // Display original matrix
    printf("\nOriginal Matrix:\n");
    for(int i = 0; i < rows; i++) {
        for(int j = 0; j < cols; j++) {
            printf("%d ", matrix[i][j]);
        }
        printf("\n");
    }

    // Display transpose matrix
    printf("\nTranspose Matrix:\n");
    for(int i = 0; i < cols; i++) {
        for(int j = 0; j < rows; j++) {
            printf("%d ", transpose[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Java Program:

```
import java.util.Scanner;


public class MatrixTranspose {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();


        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();


        int[][] matrix = new int[rows][cols];
        int[][] transpose = new int[cols][rows];


        // Input matrix
        System.out.println("Enter elements of matrix:");
        for(int i = 0; i < rows; i++) {
            for(int j = 0; j < cols; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }


        // Transpose logic
        for(int i = 0; i < rows; i++) {
```

```
            for(int j = 0; j < cols; j++) {
                transpose[j][i] = matrix[i][j];
            }
        }

        // Display original matrix
        System.out.println("\nOriginal Matrix:");
        for(int i = 0; i < rows; i++) {
            for(int j = 0; j < cols; j++) {
                System.out.print(matrix[i][j] + " ");
            }
            System.out.println();
        }

        // Display transpose matrix
        System.out.println("\nTranspose Matrix:");
        for(int i = 0; i < cols; i++) {
            for(int j = 0; j < rows; j++) {
                System.out.print(transpose[i][j] + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```